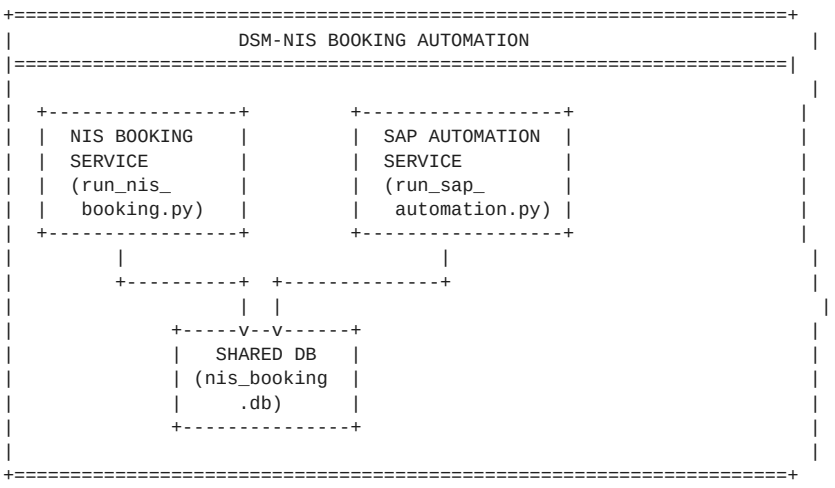
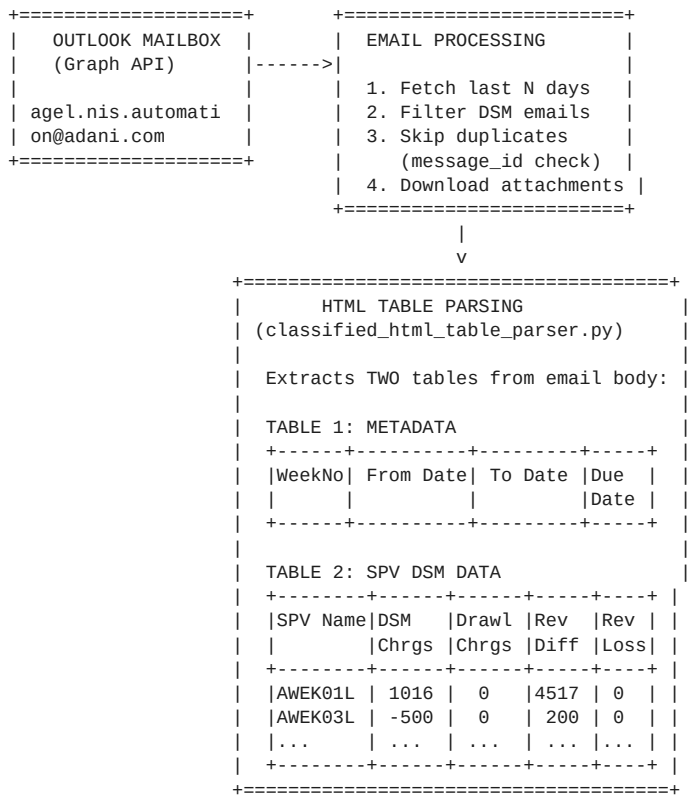
DSM-NIS Booking Automation -- Architecture & Workflow Document

1. System Overview



2. Email Ingestion & Table Extraction



Extracted Data Fields:

- Metadata: week_no, from_date, to_date, due_date, dsm_statement_published_date

- SP

3. Table Types & Row Classification

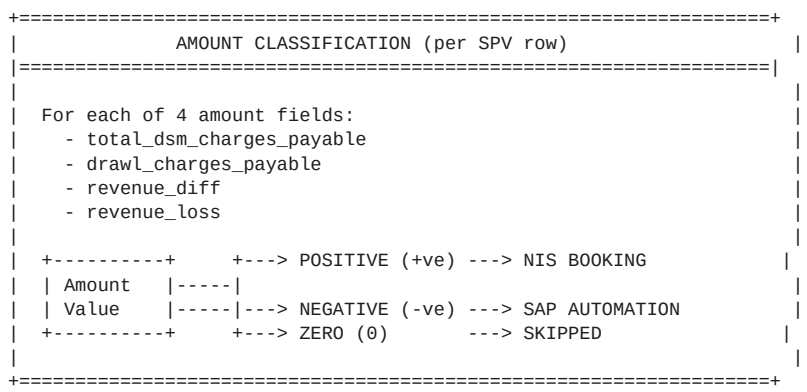
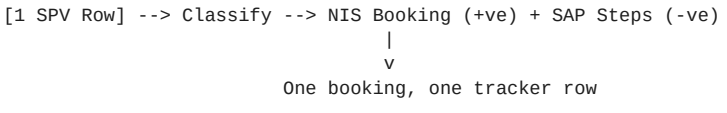


Table Types by Row Count:

Type	Rows	Behaviour
1-Row	1 SPV	Standard single booking
2-Row (Solar/Wind)	2 SPVs (same plant)	Combined into ONE NIS booking
3-Row	3 SPVs	Per-row invoice pre-extraction from PDF
5-Row	5 SPVs	Triggers Debit/Credit email to Finance

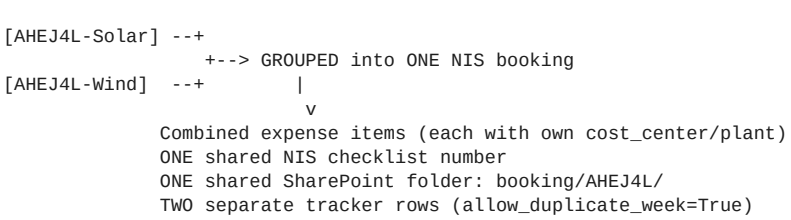
4. Table Type Handling Details

4.1 -- 1-Row Table (Standard)



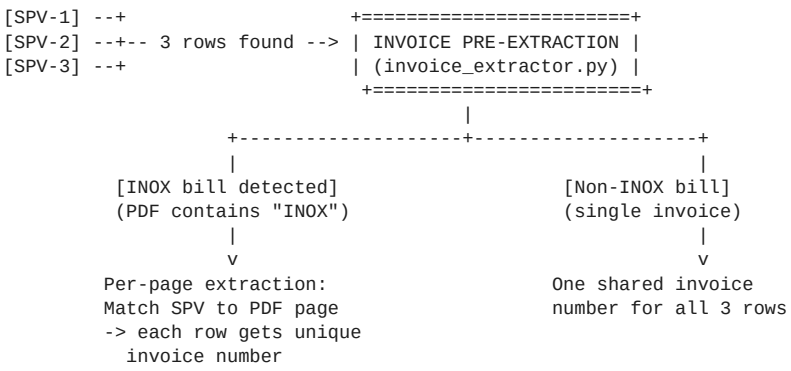
- PDF Upload: Invoice attachment PDF from email
- Vendor Data: Read from DSM-NIS-INPUT.xlsx -> "main" sheet (exact SPV name match)

4.2 -- 2-Row Table (Solar/Wind Pair)



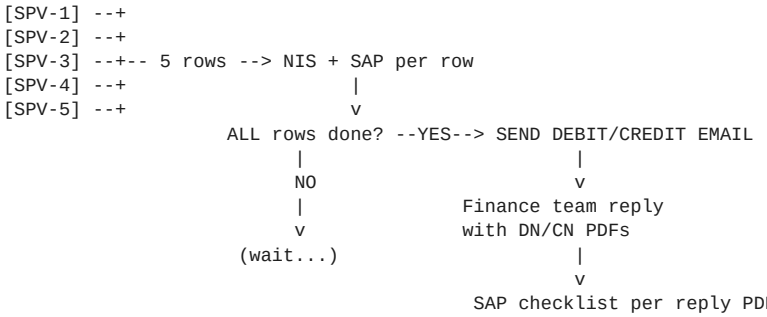
- Detection: SPV name matches regex ^\s([A-Za-z0-9]+)\s-\s(Solar|Wind)\b
- Invoice Number: From primary (first) row's vendor logic

4.3 -- 3-Row Table (Invoice Pre-Extraction)



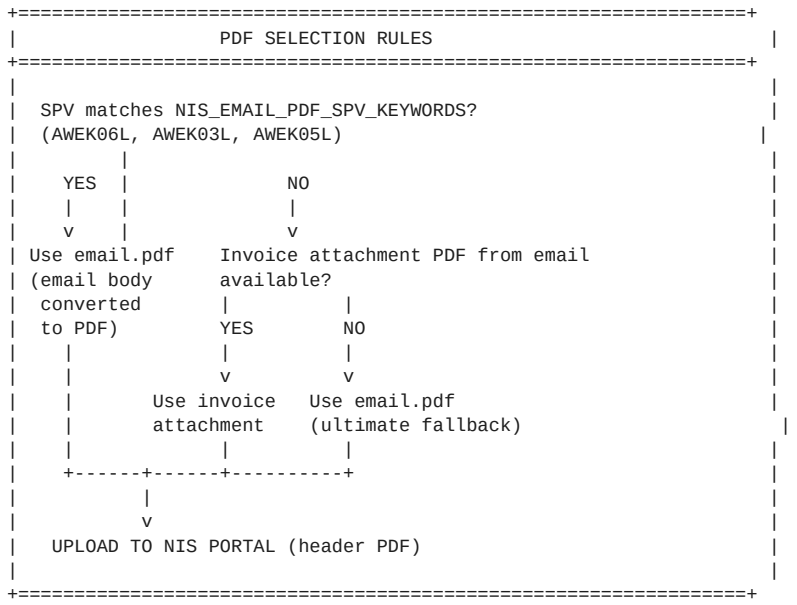
- PDF Upload: Invoice attachment PDF (each row uses same file)
- Each row booked independently with its own pre-extracted invoice number

4.4 -- 5-Row Table (Debit/Credit Trigger)



- Debit/Credit condition: len(valid_spv_rows) == config.DEBIT_CREDIT_TRIGGER_ROW_COUNT (default 5)
- Email content: Transposed HTML table with DSM Charges, Revenue Diff, Total per SPV + grand total

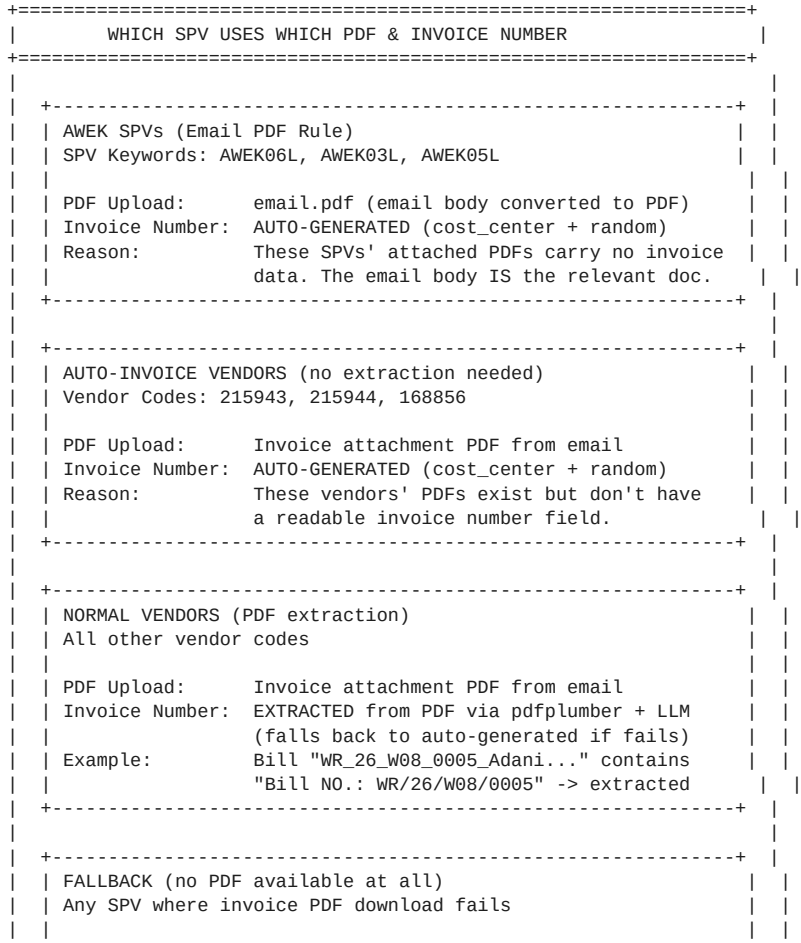
5. SPV Exceptions & PDF Rules



Invoice Number Rules:

Condition	Invoice Number Source
AWEK SPVs (NIS_EMAIL_PDF_SPV_KEYWORDS)	Always auto-generated: cost_center + ran
Vendor in NIS_AUTO_INVOICE_VENDOR_CODES	Always auto-generated
NIS_SKIP_INVOICE_EXTRACTION = True	Always auto-generated
3-row table (pre-extracted)	Pre-resolved from PDF before booking
Caller provides invoice_number	Used as-is (highest priority)
Normal vendor	Extracted from PDF via LLM (pdfplumber +
LLM extraction fails	Falls back to auto-generated

SPV-Specific PDF Upload & Invoice Rules:



```
|  | PDF Upload:      email.pdf (email body as PDF -- always       |  |
|  |                  exists in SharePoint booking folder)        |  |
|  | Invoice Number:  AUTO-GENERATED                             |  |
|  +-------------------------------------------------------------+  |
|                                                                     |
+=====================================================================+
```

Summary Table:

SPV / Vendor	PDF Uploaded to NIS	Invoice Number	Source of PDF
AWEK06L, AWEK03L, AWEK05L	email.pdf	Auto-generated	Email body -> converted to PDF
Vendor 215943, 215944, 168856	Invoice attachment	Auto-generated	Email attachment PDF
Normal vendors (all others)	Invoice attachment	Extracted from PDF (LLM)	Email attachment PDF
Any SPV (no PDF found)	email.pdf (fallback)	Auto-generated	Email body -> converted to PDF
3-row table (any vendor)	Invoice attachment	Pre-extracted per-row	Email attachment PDF

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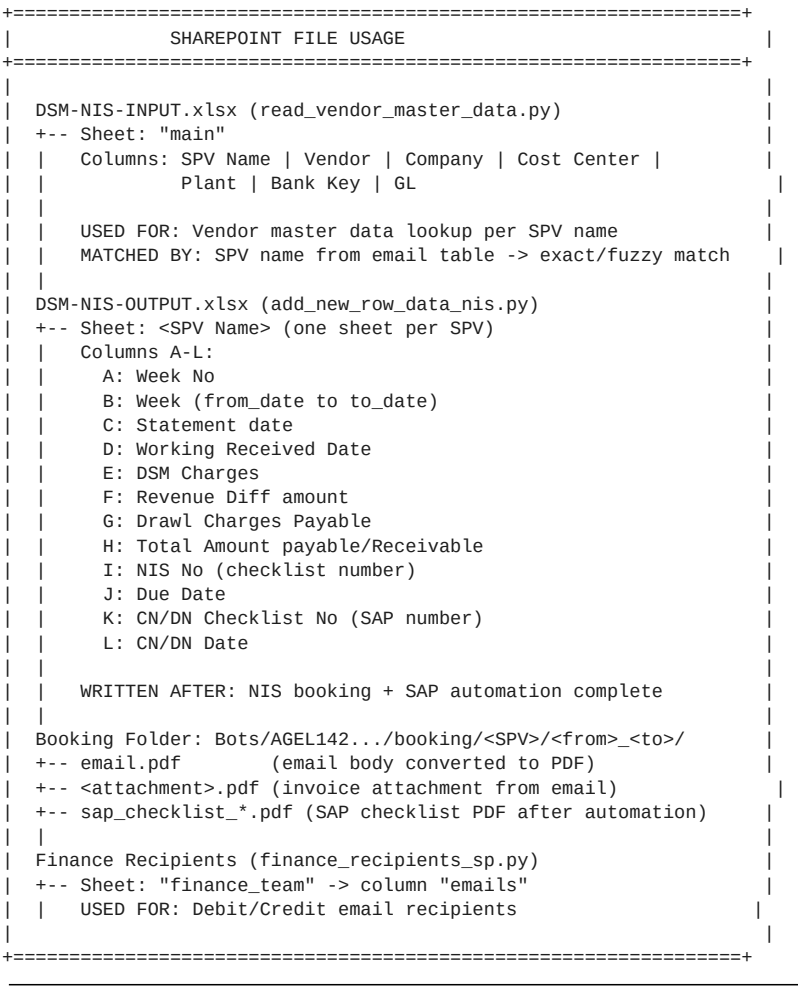
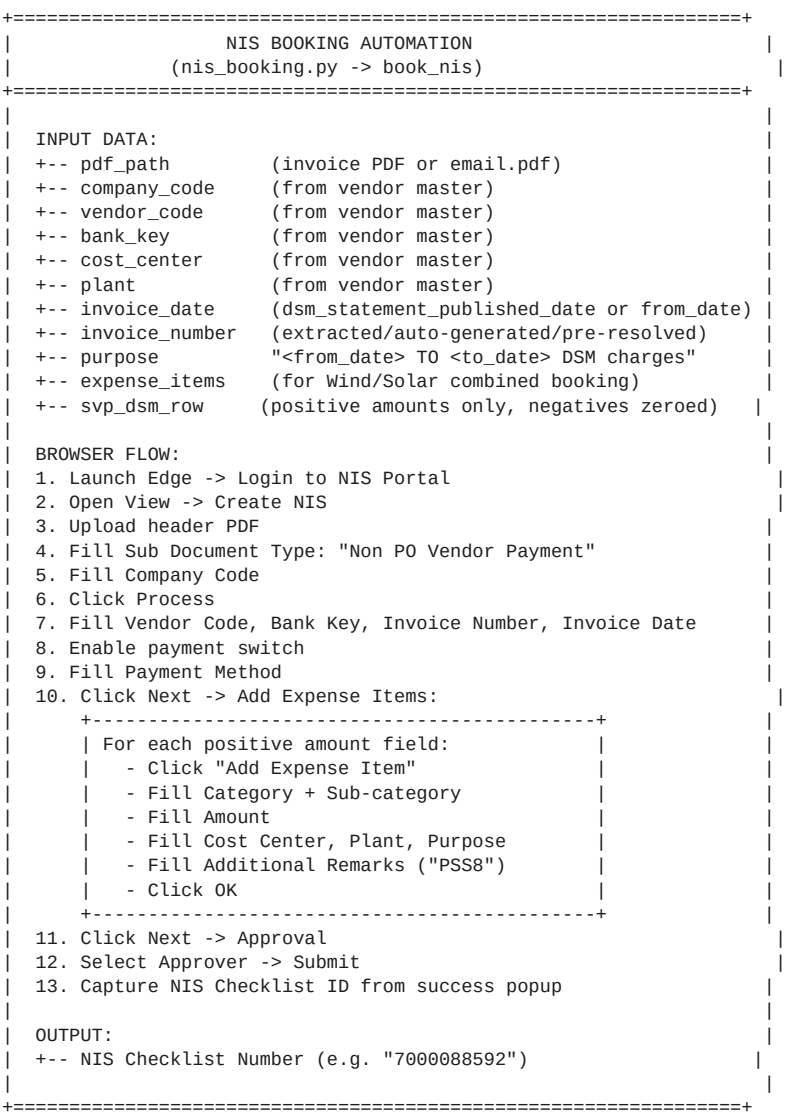
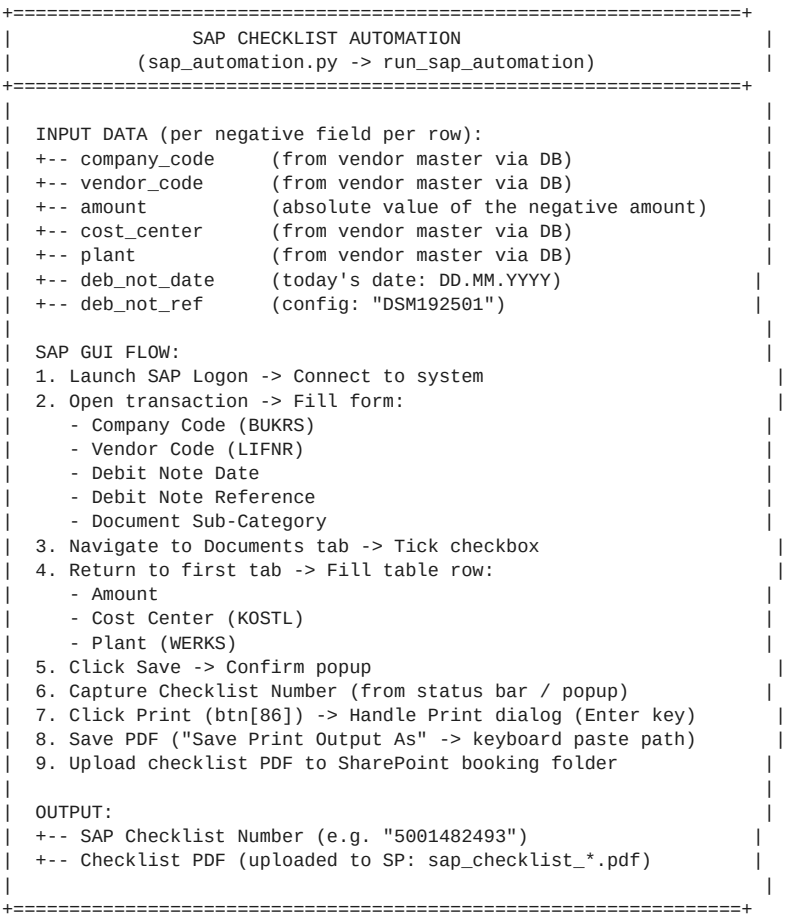
6. SharePoint Data Sources



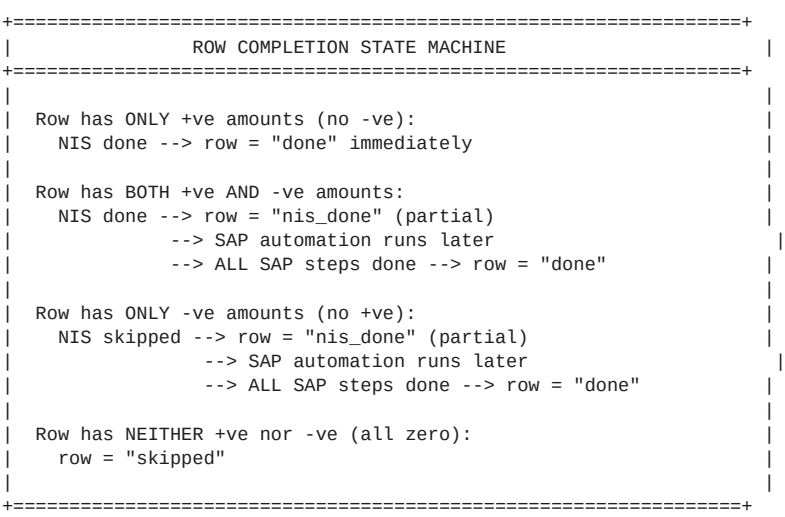
7. NIS Booking Flow (Positive Amounts)



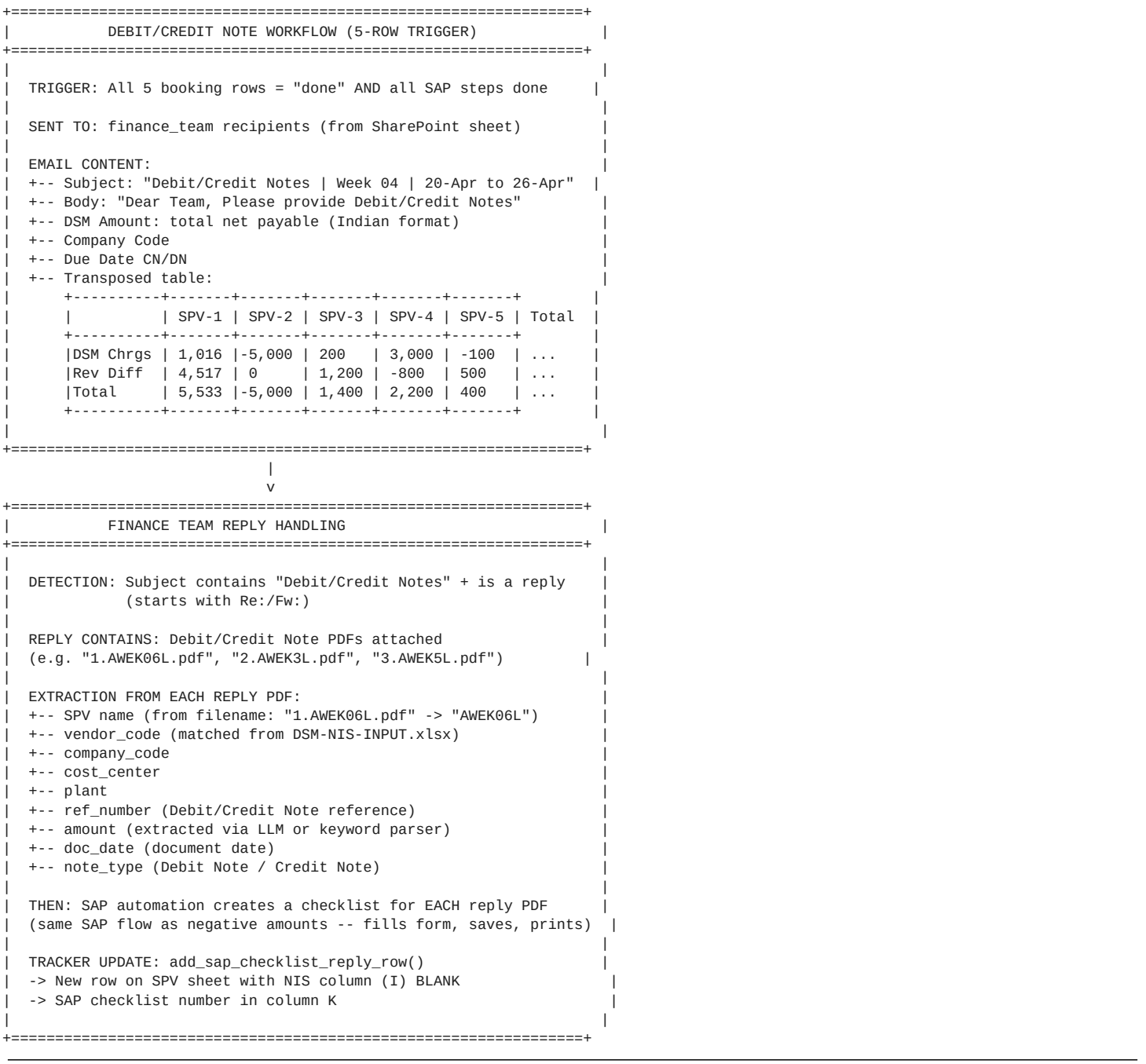
8. SAP Automation Flow (Negative Amounts)



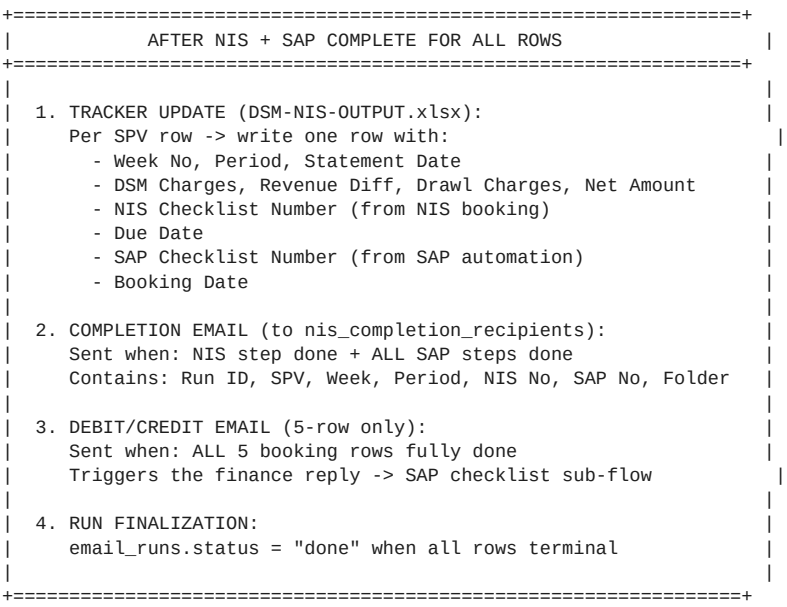
9. Row Completion Logic



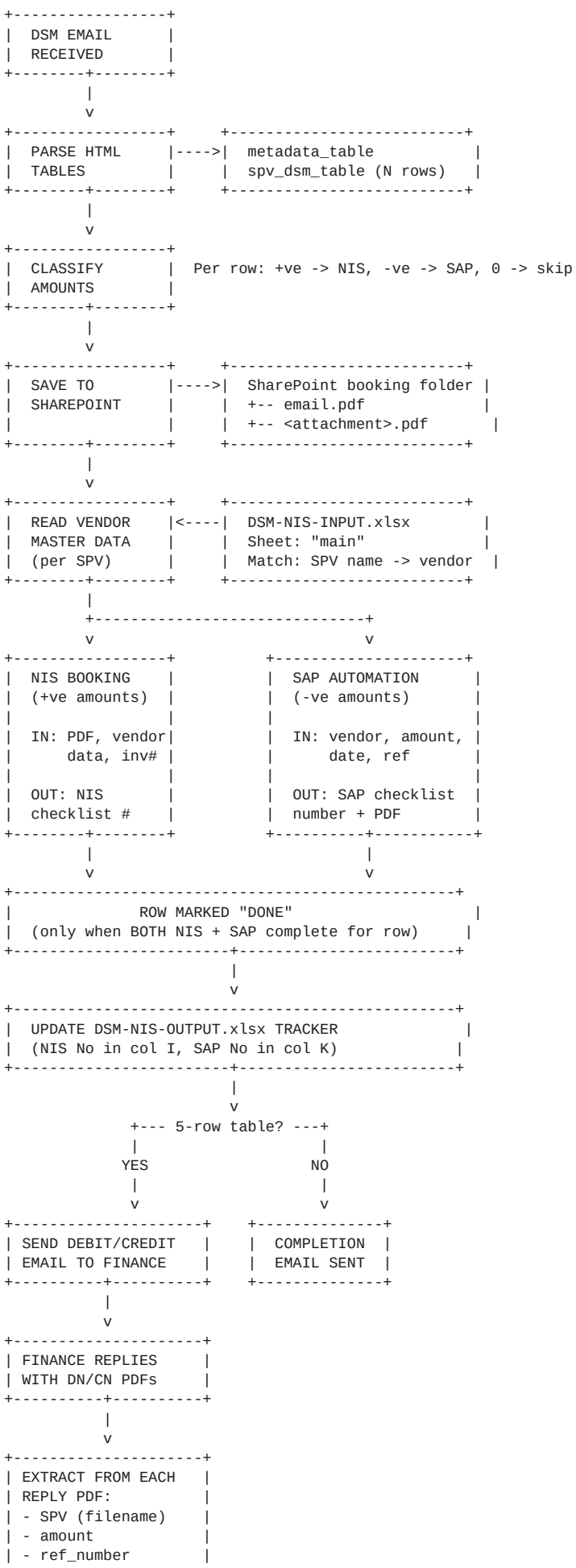
10. Debit/Credit Email Flow (5-Row Table)

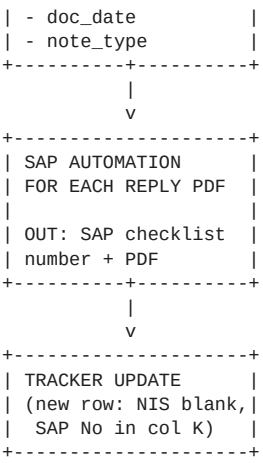


11. Post-Completion Actions



12. End-to-End Workflow (Complete)




| - doc_date |
| - note_type |
SAP AUTOMATION
FOR EACH REPLY PDF
OUT: SAP checklist
number + PDF
TRACKER UPDATE
(new row: NIS blank,
SAP No in col K)

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13. Database Schema (SQLite)

Table	Purpose
email_runs	One record per email (keyed by message_i
booking_rows	One record per SPV row (UNIQUE: message_
run_steps	Step tracking: extraction, classificatio
debit_credit_emails	Tracks sent debit/credit emails and repl
debit_credit_pdfs	One record per finance-reply PDF -> SAP

Key Status Values:

- pending -> waiting to be processed
- nis_done -> NIS complete, awaiting SAP
- failed -> error occurred

- ru
- do
- sk

14. File & Module Map

File	Role
run_nis_booking.py	Entry point: NIS service
run_sap_automation.py	Entry point: SAP service
nis_booking/main.py	NIS orchestrator: email scan -> prepare
nis_booking/nis_booking.py	Browser automation: NIS portal form fill
nis_booking/db_manager.py	SQLite database layer (shared by both se
nis_booking/classified_html_table_parser	Email HTML -> structured tables
nis_booking/invoice_extractor.py	PDF -> invoice number (pdfplumber + LLM)
nis_booking/read_vendor_master_data.py	SharePoint Excel -> vendor data
nis_booking/add_new_row_data_nis.py	Write tracker rows to DSM-NIS-OUTPUT.xls
nis_booking/debit_credit_email.py	Build debit/credit email HTML
nis_booking/sharepoint_upload.py	Upload/download files to/from SharePoint
sap_automation/sap_main.py	SAP orchestrator: find pending tasks ->
sap_automation/sap_automation.py	SAP GUI automation: form fill -> save ->
sap_automation/sap_config.py	SAP-specific configuration